Discuss and give your answer
You know that plant produce their food by a process called photosynthesis.
Can we call photosynthesis a chemical change?

3.4.1 Rusting of iron

In class six, we have already studied that rusting is an example of a chemical change. Now, shall we read why the process of rusting is called a chemical change.

The Iron Pillar at Delhi

Amazingly there is an iron that did not rust!



There is an iron pillar at the Qutub Minar in Delhi which is more than 1600 years age. Even after such a long period, the iron pillar kept in open spaces has not rusted at all. This shows that Indian scientists made great advances in metal making technology even at 16th century which enabled them to make this iron pillar having the quality of great rust resistance.

Rusting is one change that affects iron articles and slowly destroys them. Since iron is used in making bridges, ships, cars, truck bodies and many other articles, the monetary loss due to rusting is huge. The process of forming rust is represented as follows:

iron + oxygen + water→rust

$2Fe + 2O_2$ from air + $2H_2O \longrightarrow 2Fe_2O_3 . H_2O$

For rusting to take place both oxygen and water (or even water vapour) is essential. In fact, if the content of moisture in air is high, the air is said to be more humid and eventually rusting is faster.

How can we prevent rusting?

Iron articles can be prevented from making contact with oxygen, water/water vapour. A simple way is to apply a coat of paint or grease. These coats should be applied regularly to prevent rusting.

Another way of preventing rusting is to deposit a layer of a metal like chromium or zinc on iron. This is called galvanization and you will learn about this detail in higher classes.

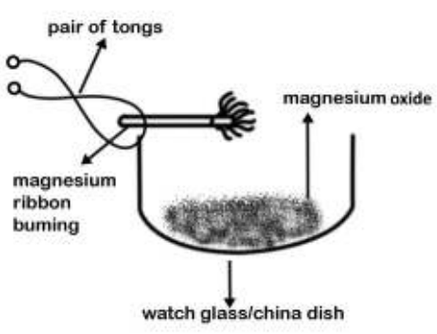


3.4.2 Burning

we have already studied that burning of paper is a fast change. Burning a piece of paper gives entirely new substances such as carbon-di-oxide, water, water vapour, smoke and ash. Heat and light are also given out during the burning of

ACTIVITY 7

Take a small piece of magnesium ribbon and clean it by rubbing its surface with a sand paper. Hold the magnesium ribbon at one end with a pair of tongs and bring its other end over the flame of a burner.



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